



**BHASKARACHARYA NATIONAL INSTITUTE FOR
SPACE APPLICATIONS AND GEO-INFORMATICS**
WEEKLY PROGRESS REPORT (08/06/2026 – 14/06/2026)

WEEK 4

PROJECT NAME	Prithvi EO Models (IBM Model)
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PROJECT DESCRIPTION :

Fine tuning and Training of IBM Prithvi EO Models for Earth Observation tasks.

GROUP MEMBER :

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GROUP ID :

2026G264

GROUP GUIDE :

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COLLEGE NAME :

Gati Shakti Vishwavidyalaya, Vadodara

08/05/2026	• Refined the live execution pipeline status tracking. • displaying live checkbox status in the run segmentation pipeline.
09/05/2026	• changing and updating the UI for the Prithvi land classification.
10/05/2026	• Add a comparison module allowing users to load two historical layers (from different dates) from the database and compute pixel-wise change detection.
11/05/2026	• Develop a color picker UI in Streamlit that dynamically generates a Styled Layer Descriptor (SLD) XML file.
12/05/2026	• Making request to geoserver to view output layers instead of reading it from local cog file.
13/05/2026	Holiday (2nd Saturday)
14/05/2026	Holiday (Sunday)

Next Week Plan	Preparing and Verificaion of Report, presenting our project.
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REFERENCE:

- [ibm-nasa-geospatial \(IBM-NASA Prithvi Models Family\)](#)
- <https://www.youtube.com/live/CB3FKtmuPI8?si=olB4EmCtkUjSo0r>
- [Sentinel-2 | Copernicus Data Space Ecosystem](#)
- <https://research.ibm.com/blog/terramind-prithvi-tiny-small-models-geospatial>
- <https://github.com/NASA-IMPACT/Prithvi-EO-2.0>

SCREEN SHOTS:

Navigation

- Run Segmentation
- Saved Layers History

Connection & Info

Backend API URL

`http://localhost:8001`

API Server Online
Model: prithvi_eo_v2_300_t1
GPU Active

Prithvi-EO Land Segmentation

Powered by NASA-IBM Prithvi-EO-300M Foundation Model. Select raw satellite archives or bands and perform pixel-level classification.

Step 1: Browse & Select Server Files

Input Folder or File Path (on server)

`"C:\Users\chsid\Downloads\S2B_MSIL2A_20260519T045659_N0512_R119_T44PLU_20260519T084811.SAFE.zip"`

Browse Server Filesystem

Step 2: Selected Data Verification

Selected Path on Server:

True Color Composite vs Segmentation Output

Left -- Input satellite imagery as a True Color Composite (Red→R, Green→G, Blue→B). Looks exactly like Google Maps / real photography. Right -- Pixel-level classification output from Prithvi-EO-300M.

Input -- True Color Composite

Output -- Segmentation Map

Navigation

Run Segmentation

Saved Layers History

Connection & Info

Backend API URL

http://localhost:8001

API Server Online

Model: prithvi_eo_v2_300_tl

GPU Active

Saved Layers History Repository

Browse previously processed segmentation files, compute real-time spatial calculations, and retrieve WMS links for GIS tool integrations.

Class 1 Label

Vegetation

Class 2 Label

Urban

Class 3 Label

Water

Class 4 Label

Barren

Database Historical Records

Refresh Database Listings

☐ Also delete COG file from disk when removing ⓘ

Layer Name	COG Path / Created At	GeoServer	Load	Delete
ahmedabad_testing	Date: 2026-06-11 16:55	Local	Load	Delete